

# From documentation chaos to a searchable document server for small teams

A personal documentation mess — emails, Joplin, WhatsApp, dev.to posts — became Doclets: a full-stack AllSpeak app with content search, local-LLM queries, and per-topic access control.

A while back I wrote that I produce a lot of documentation but had no strategy for organising it. Emails, web pages, Markdown notes in Joplin, WhatsApp messages, dev.to posts of my own — it was a largely unsearchable mass of information. Every fact I needed lived somewhere; none of it was findable.

Doclets started as an AllSpeak coding exercise. It became the tool that brought order to that chaos — and, six months on, it has quietly turned into something a small team could genuinely use.

## What Doclets is

---

Doclets is a central, searchable repository of Markdown documents ("doclets"), organised into topics and years. You read and edit doclets in a browser on any device; a small server process owns the documents and talks to the browser over MQTT.



Screenshot: the Doclets reader showing the topic list and query bar

The two things that make it more than a notes folder are *finding* and *sharing*.

## Finding things

---

Plain search is a simple substring match over every doclet's title and content — nothing clever, but it works, and it already beat the alternative (not being able to find things at all).

More recently I added a second way to search: a local language model. Tick "LLM query" and you can ask questions like *"list the main topics covered by the doclets in the Linux topic"* and get a short prose answer derived from the collection's subjects, or *"find the doclets that include a proposed letter of introduction"* and get the matching documents back. The model runs on your own machine via Ollama — no cloud, no per-query cost — and every doclet is indexed by a small embedding model, so searching doesn't mean reading the whole database on each query.

 Screenshot: an LLM query returning a concise answer

## Sharing with a small team

---

None of the tools that held my scattered notes offered the same ease of use — and few could operate across the internet. Doclets is a central repository, so the same collection is reachable from any browser, anywhere.

If you want to share that repository with colleagues, each topic can have an owner and a visibility:

- **public** topics are readable by anyone with the link
- **private** topics are readable only by the owner and named readers
- creating/modifying is controlled by per-user grants, deleting by its own separate grant
- a simple activity log records who did what, and when

Identity is a token phrase — nothing to install, nothing to configure beyond a small JSON file of permissions.

 Screenshot: the per-topic access control file

## Why the construction matters

---

Doclets is also a good illustration of building a real application in AllSpeak, and the way it is put together is one of the reasons it stayed maintainable:

- **One language, full stack.** The browser client ( `doclets.as` ) and the server ( `docletServer.as` ) are both AllSpeak. One person can understand and change the entire system — UI, messaging, backend — without a conventional web framework or a second client language.
- **Declarative screens.** The UI is defined as Webson JSON, not hand-written DOM code — easy to read, easy for AI tools to generate.
- **MQTT instead of REST.** The browser and server talk over MQTT request/reply, which works naturally across the internet and on phones.
- **A clear plugin boundary.** The one genuinely heavy part — managing, searching and securing the document collection, plus the LLM integration — lives in a small Python plugin that exposes simple AllSpeak commands like `doclets query` and `doclets topics` . The scripting language stays readable; the native code stays isolated.
- **Features that arrive as additions, not rewrites.** Semantic LLM search and per-topic access control were bolted on over time without disturbing the original design — a sign

the structure is doing its job.

Because AllSpeak separates language from logic, the same application could be presented in French, Arabic or any other language via a language pack — a property that also happens to be at the heart of why AllSpeak exists.

## Running it

---

The server is a single Python process (the `allspeak` runtime) plus an MQTT broker; the client is static files on any web host. If you want the LLM features, you add Ollama with a small model — again, all local.



Screenshot: the doclets server console showing MQTT connect and an LLM query

Six months ago my documentation was a scattered mess. Today I can find any of it in seconds, and I can share the collection with people who need it without giving them the mess. If you have the same problem, Doclets is a good place to start — and if you are curious about AllSpeak, it is a working example of what the platform can do.